

Adversarial Review for Market Convergence: A Cross-Paper Analysis

Q Competitive Market Behavior of LLMs [arXiv:2609.02580](#)

P Adversarial Review: Structured Disagreement for Grounded Agentic Code Review
[arXiv:2608.18167](#)

Research phase: PAUSE-ON-ITERATE after 1 round, 1 ITERATE, 0 REVISE

The Pair

Q examines LLM agents in double-auction markets, finding slower or absent convergence to equilibrium compared to human traders, with substantial heterogeneity across model families and market roles. CoT analysis reveals trading decisions correlate with a shift from strategic considerations toward urgency [\[ledger 1, 3\]](#).

P introduces Adversarial Review (AR), a three-agent protocol where a reviewer evaluates code and a critic audits through structured disagreement before the main agent edits. AR outperforms larger multi-agent baselines on code-review tasks, specifically addressing false-consensus failure modes [\[ledger 1, 3\]](#).

Connexion Pursued

Initial hypothesis: P's structured disagreement protocol could improve Q's market convergence. The proposed mechanism was that AR-style reviewer-critic friction might interrupt premature trade commitments, forcing evidence-grounded justification before execution [\[ledger 1, 3\]](#).

Supported Findings

The peers established:

- Q's core pathology is *heterogeneity* (agents failing to coordinate), not false-consensus. These are opposite failure modes [\[ledger 2, 5\]](#).
- Q's CoT finding (strategic→urgency shift) operates at the *individual* agent level, while heterogeneity is a *market-level* phenomenon [\[ledger 6\]](#).
- AR is deliberative and collaborative; double-auction trading is competitive and time-sensitive. Adding disagreement overhead could worsen convergence by introducing latency [\[ledger 4\]](#).

- A narrow salvageable connection exists only if: (1) urgency-driven trades are insufficiently justified at the individual level, (2) self-critique reduces decision noise, and (3) this improves market-level coordination [\[ledger 6\]](#).

Objections

1. **Pathology mismatch:** Q shows heterogeneity (insufficient agreement); P addresses false-consensus (excessive agreement without evidence). These are opposite pathologies [\[ledger 2, 5\]](#).
2. **Domain mismatch:** Code review tolerates latency; double auctions are time-sensitive. AR friction could harm rather than help [\[ledger 4\]](#).
3. **Role mapping unclear:** P's reviewer/critic roles have no obvious analogue in trading. What counts as "evidence" for a trade remains undefined [\[ledger 6\]](#).

Unresolved Issues

Evidence is thin on:

- Whether urgency-driven trades are the *cause* of non-convergence or a symptom of deeper misunderstanding of market dynamics [\[ledger 4\]](#).
- Whether heterogeneity is structural (different models/roles inherently incompatible) or a result of correctable decision-making patterns [\[ledger 5\]](#).
- What a trading-adapted AR protocol would look like: timing, evidence standards, and whether self-critique (vs. multi-agent) suffices [\[ledger 6\]](#).

Smallest Next Step

Examine Q's full CoT traces to determine whether individual agents exhibit premature commitment patterns that resemble false-consensus at the decision level (before trade execution), even while market-level outcomes show heterogeneity. This would establish whether AR-style interruption is mechanistically plausible.